

Name of the Module : JSE (Java Standard Edition) [Core Java]

Sessions : 105 session

Version : Java 25V [LTS, Long Term Support]

Notes : Google classroom (PDF format)

Assignments : classroom.coderide.com + lab.coderide.com + devride.com

Lab : Lab is compulsory

FAQ : All FAQ (600+ FAQ)

3+years : Developers Knowledge

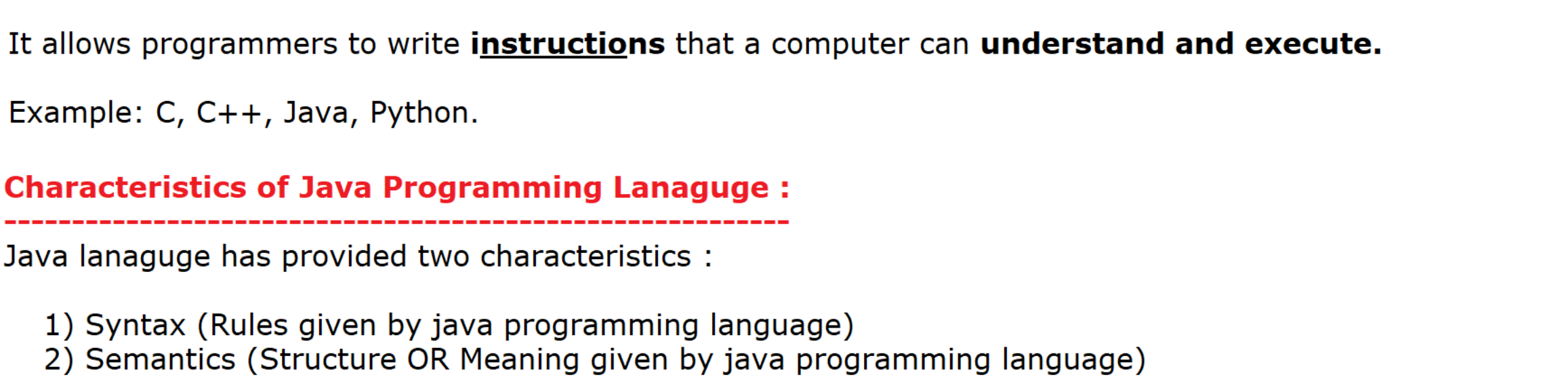
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What is a Language ?

A language is a medium of communication through which we can express our thoughts and ideas and communicate with others.

Example: English, Telugu, Hindi, French, etc.

What is Programming language?



A programming language is a **medium of communication** between a programmer and a computer.

It allows programmers to write **instructions** that a computer can **understand and execute**.

Example: C, C++, Java, Python.

Characteristics of Java Programming Lanaguge :

Java lanaguge has provided two characteristics :

- 1) Syntax (Rules given by java programming language)
- 2) Semantics (Structure OR Meaning given by java programming language)

Syntax :

Every Programming language has provided some syntax so, developer should follow while writing the code.

Example :

Subject + Verb + Object [English Grammar Rule]

He is a boy. //Valid

He is a box. //Invalid

In the syntax part our compiler will only verify the syntax of the code, It will never verify whether the code is meaningful OR meaning less.

Example :

```
int x = 10;
```

```
int y = 0;
```

```
int z = x/y;
```

Note : The above code will compile without any problem but at runtime our runtime environment will generate runtime error.

Semantics :

In Semantics, Our runtime environment will verify whether the code is meaningful or not, It will verify the structure, If everthing is correct then only It will execute the code.

So the final conclusion is, Our Java code will be verified by two times :

a) At the time of compilation [Syntax Verification]

b) At the time of execution [Semantics Verification]

What is Java ?

Java is high level, platform independent, robust(Strong), multi-threaded,object-oriented and secure programmimng language.

It was developed by sun microsystem in the year 1995.

The first version of java was released on 23rd January 1996.

In the year 2010, Java was overtaken by Oracle Corporation.